SULIT

SCHOOL OF COMPUTING
Faculty of Engineering

UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER II 2021/2022

SUBJECT CODE	**: SECR2043**
SUBJECT NAME	**: OPERATING SYSTEM**
SECTION	**: SECR/J/B/V/P/H/ SCSR/J/B/V/P/H**
TIME	**: PAPER 2 – 90 MINUTES**
DATE/DAY	**: 18 JULY 2022 (MONDAY)**
VENUES	**: Online**

INSTRUCTIONS:

Answer all questions from Paper 1 and Paper 2. Show all your works.

This test will contribute 35% towards the total marks of 100%.

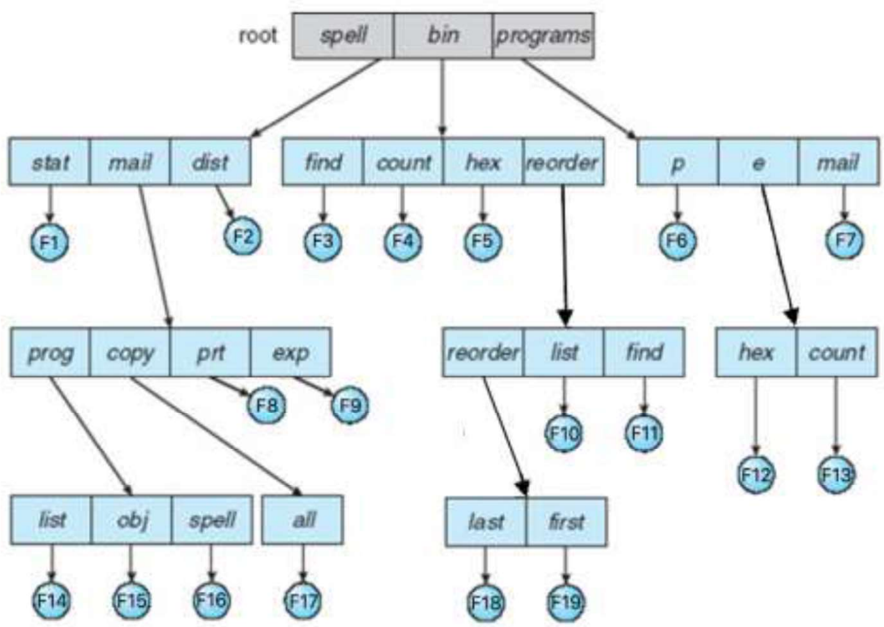
(Please Write Your Lecturer Name and Section in Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section (Circle)	**01 / 02 / 03 / 04 / 05 / 06 / 07 / 08 / 09 / 15**
Lecturer Name (Circle)	**Dr Zuriahati / Mrs Rashidah** **Dr Nur Haliza / Dr Nor Shahida**

This question paper consists of 4 printed pages excluding this page.

QUESTION 3 [15 MARKS])

a) Refer to the diagram below, assume the `LINUX` command prompt environment is used and answer the following questions. Show all your work.



i. Determine the absolute path name for file, `F6` and `F19`. [2]

ii. Determine the relative path name for file, `F7` and `F17`. [2]

iii. Identify the current directory for file, `F8`. [1]

iv. A filename without the path information can also be called as: [1]

v. File `F16` has a subdirectory with 9 protection bits, 3 for owner, 3 for group and 3 for public. Each class is assigned with **R**, **W** and **D**. Following the value (in decimal) assigned to the class, explain the access permission for each of the classes. [3]

`Owner = 6, group = 5, public = 1`

vi. File `F16` later changed using this command "`chmod 743 F16` " (for access read, write, execute), write the `UNIX` directory for the file `F16`. [2]

vii. Explain briefly what is the explanation for the `UNIX` directory given. [2]

-rwxrwxr-x 7 research admin 512 Jul 16 01.12 public.doc

drwx-wxr-x 2 research librarian 1024 Aug 31 03.23 public/

viii. In the new directory, write <u>**TWO**</u> possible single command lines to set an access control of the `.doc` file to be classes as read only. [2]

QUESTION 4 [20 MARKS])

Consider the reference string:

4	3	2	1	2	3	4	7	3	2	1	2	4	5	1	2	1	3	2	1

a) How many page faults if we have 4 frames using FIFO? Show all your work. [4]

b) How many page faults if we have 3 frames using FIFO? Show all your work. [4]

c) From the results above, perform the comparative analysis and discuss about system efficiency related to the page faults amount and the frame number used for FIFO. [4]

d) How many page faults if we have 4 frames using OPT? [4]

e) From the results above, perform the comparative analysis and discuss about system efficiency related the page faults that related with the different algorithms used. [4]

QUESTION 5 [15 MARKS])

a) By using the C-SCAN disk scheduling algorithms, illustrated by drawing the movement of the disk arm to satisfy the requests. Label clearly. [5]

✓ A disk with 160 cylinders.
✓ A request queue:
97, 100, 50, 134, 26, 145, 80, 90
✓ Head pointer 75.
✓ Direction: the disk arm is moving to innermost

b) Calculate the total distance (in cylinders) that the disk arm moves to satisfy the requests. Show all your work. [2]

c) If the SCAN disk scheduling algorithm is used, briefly state a conclusion that can be made about the disk arm movement to satisfy the requests. [3]

d) Using similar details from question (a), redraw the movement of the disk arm using SCAN algorithm with direction of the disk arm moving toward 0. Label clearly. [5]